1. Immunity

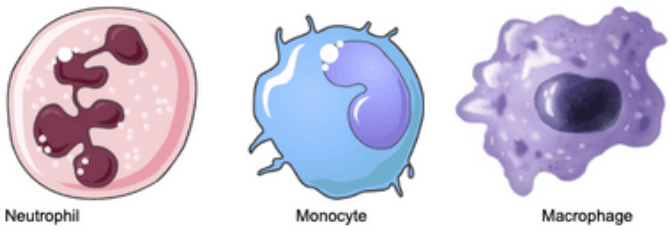
1.1. The Immune System

Immune response:

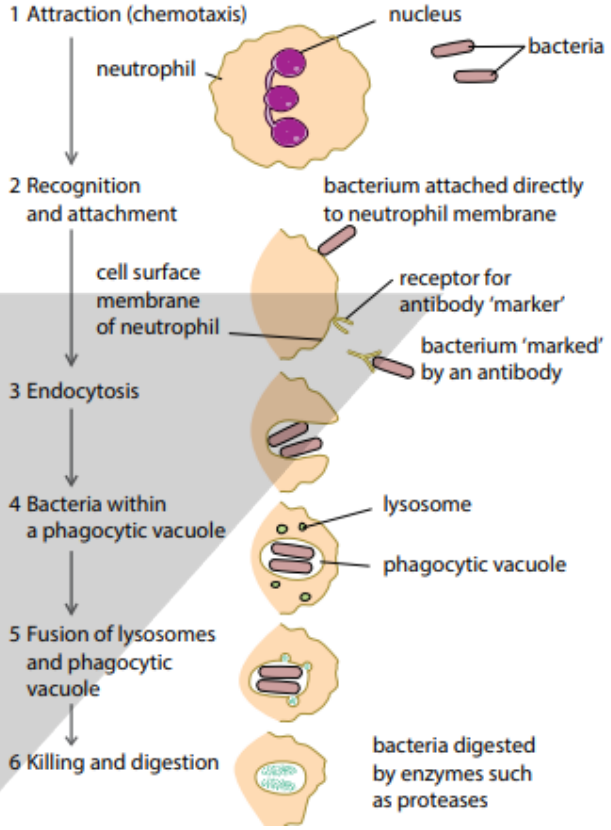
- The body's immune system responds to non-self-antigens by lymphocytes and phagocytes (the two main kinds of white blood cells)
- It involves the production of antibodies and the killing of cells that have become infected by pathogens
- Non-self antigens are substances or cells recognised by the immune system as being foreign and hence, stimulate an immune response
- Self-antigens are substances produced by the body that the immune system does not recognise as foreign and, hence do not stimulate an immune response

Phagocytes

- Phagocytes:
  - Produced in and stored in the bone marrow throughout life before being distributed around the body by blood
  - They remove dead cells as well as invasive microorganisms
    - Neutrophils:
      - Most abundant WBC
      - They circulate in the blood and leave through capillary walls to enter tissues during an infection.
      - They are short-lived and often die after ingesting and destroying bacteria, forming pus.
    - Monocytes → (mature to) macrophages
      - Made in bone marrow
      - Are long-lived and larger than neutrophils, they are less abundant
      - Monocytes in blood; macrophages found in lungs, liver, spleen, kidney & lymph nodes where they engulf foreign particles and microorganisms
      - Also known as antigen-presenting cells (APCs), they ingest antigens and display them on their surface, allowing T-lymphocytes to bind to the antigen and stimulate immune response
      - They cut up pathogens using lysozymes



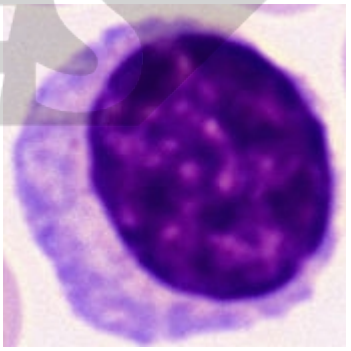
- Phagocytosis:
  - During an attack by pathogens, cells produce chemicals called histamine, which attract neutrophils to the site of infection (chemotaxis), after which the following steps take place:



1.2. B and T Lymphocytes

Lymphocytes:

- WBC produced before birth in the bone marrow
- They are smaller than phagocytes and have a nucleus that fills most of the cell



B-lymphocytes:

- Produced and mature in the bone marrow, concentrating in lymph nodes and the spleen
- When mature, each B cell produces one type of antibody molecule
- Part of the antibody molecules form a glycoprotein receptor that combines specifically with one type of antigen

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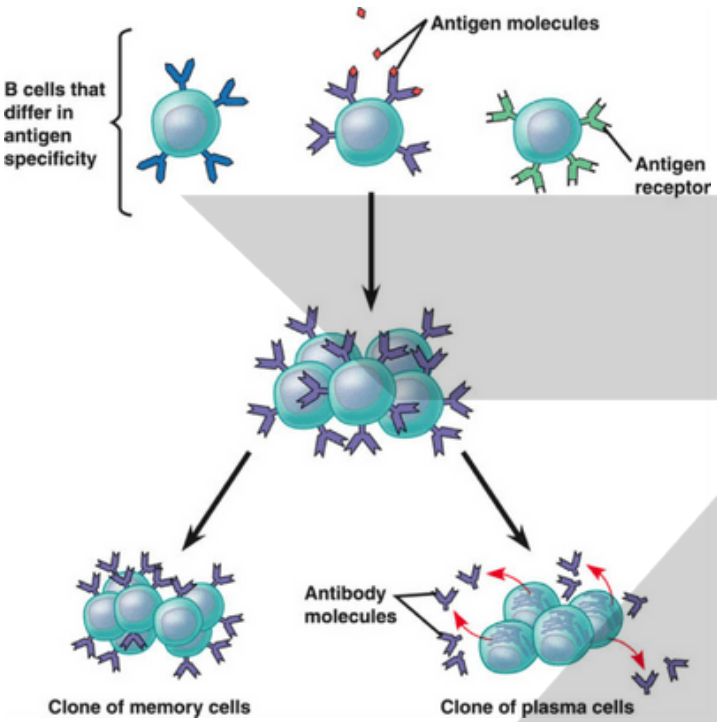
- Once combined, the B cell will undergo mitosis repeatedly to form clone cells
- Only the B cells with antibodies complementary to antigens divide like this, known as clonal selection, followed by clonal expansion
- Some differentiate into short-lived plasma cells (release antibodies into blood/lymph/linings of lungs and gut), and others become long-term memory cells (have glycoprotein receptors on the cell membrane)



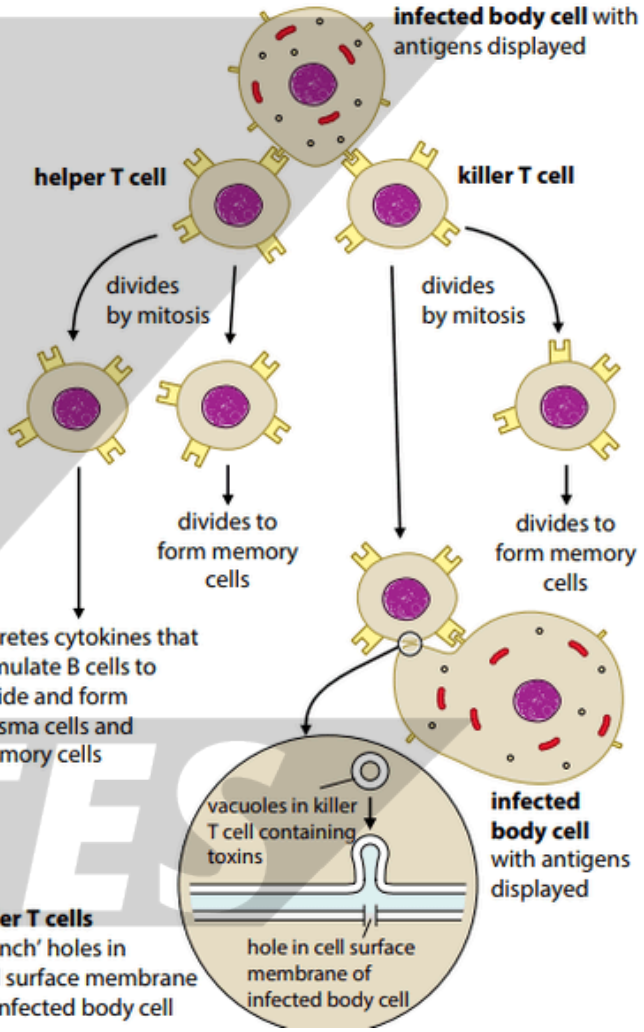
Plasma cell:

- There is an extensive network of RER in its cytoplasm for the production of antibody molecules, which plasma cells secrete into blood or lymph by exocytosis
- The mitochondria provide ATP for protein synthesis and the movement of secretory vesicles.

T-lymphocytes:

- These cells are produced in the bone marrow but collect in the thymus till maturity
- Mature T cells have receptors that are complementary to antigens of pathogens
- T cells only respond when they encounter this antigen on the cell surface membrane of a host cell, such as a macrophage (APC)
- A particular T cell with a complementary receptor will bind to an antigen found on the surface of APC

- Clonal selection and expansion also take place and differentiate into the following:
  - T helper cells secrete hormones called cytokines, stimulating B and T killer cells to divide and macrophages to carry out phagocytosis more vigorously
  - T killer cells/ cytotoxic T cells destroy the cell to which they are bound
  - They search the body for cells that have become invaded by pathogens and are displaying the pathogen's antigen on their plasma membranes.
  - When T-killer cells recognise the antigens, they attach themselves to the surface of infected cells and secrete toxic substances that kill the cells and pathogens within them.



1.3. Primary and Secondary Immune Response

The number of WBCs increases at the time of infection:

- Neutrophils during bacterial infection and whenever tissues become inflamed and die
- Lymphocytes in the blood increase in viral infections and TB
- Most of the lymphocytes that circulate in the blood are T cells

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- HIV invades helper T cells and causes their destruction, so blood tests for people who are HIV+ record the numbers of specific T cells

Memory cells:

- Form the basis for immunological memory lasting many years, often a lifetime and are involved in long-term immunity

Primary response:

- Very few B cells are specific to the antigen; thus, antibodies are produced at low

Secondary response:

- Many more antibodies are produced as many memory cells divide quickly and differentiate into plasma cells

1.4. Antibodies

- Globular glycoproteins with quaternary structure, forming plasma proteins called immunoglobulins
- 4 polypeptide chains, 2 heavy chains and 2 light chains which are held together by disulphide bridges and form a Y-shaped structure

Variable region:

- Upper part of the 'Y'
- Provides 2 identical antigen binding sites
- Specific for binding antigen as it is **complementary** shape to the antigen
- R groups at antigen binding site forms H-bonds and ionic bonds with specific antigen
- Primary structure at var. region is different for each type of antibody

Constant region:

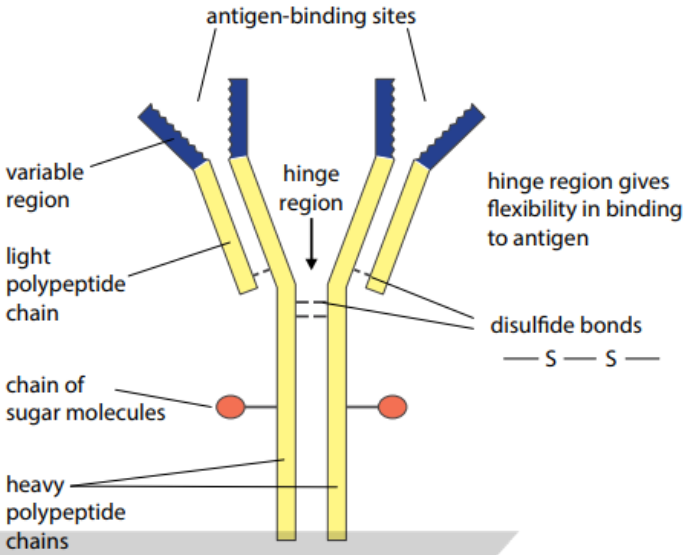
- Formed by heavy and light chains - lower part of the 'Y'
- when circulating in the blood, they **bind to receptors on phagocytes**
- Divides up antibody into classes eg. igM, igG, igA, igE

Hinge region:

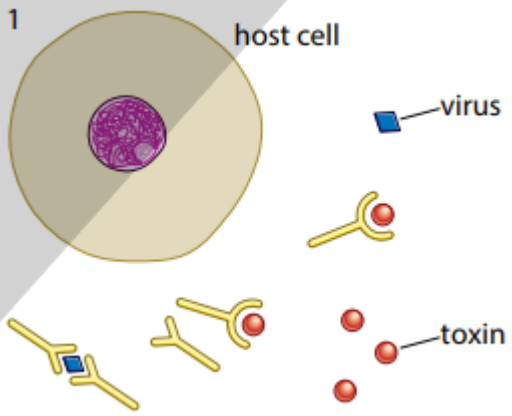
- Held by disulfide bridges
- Gives flexibility when binding to antigen

Specificity of antibodies:

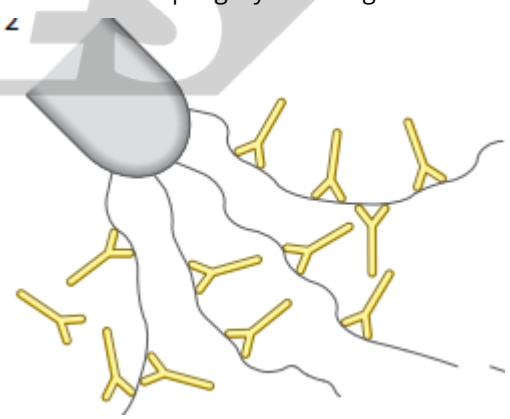
- The sequence of amino acids in this region make the specific 3-D shape which binds to just one type of antigen
- R group interactions with the antigen gives it the specific shape



- Antibodies protect the body from pathogens in various ways such as:



- Combine with viruses and bacterial toxins preventing them entering or damaging cells
- Antibodies that combine with toxins and neutralize them (antitoxins)
- Attach to the flagellum of the bacterium making them less active and easier for phagocytes to engulf



- Agglutination (clumping together) of bacteria reducing the chances of spread throughout the body

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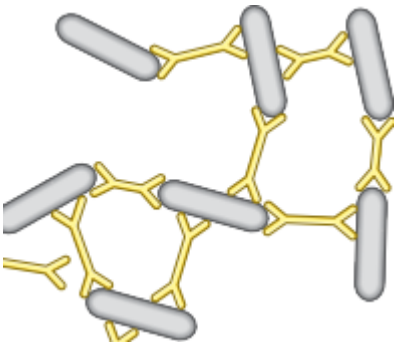
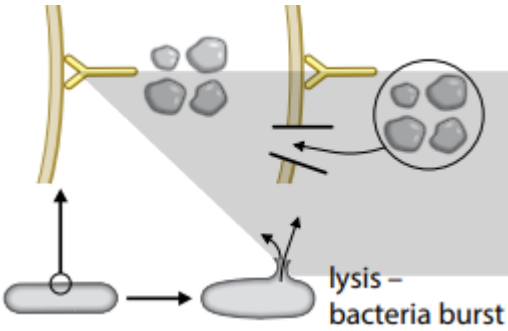
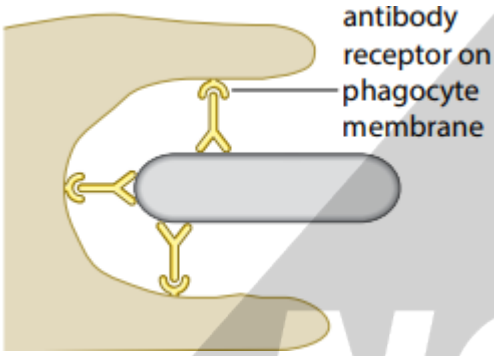


- Punch holes in the cell wall of bacteria, causing them to burst when they absorb water by osmosis



- Antibodies coat bacteria, making it easier for phagocytes to ingest them; phagocytes have receptor proteins for the heavy polypeptide chains of antibodies



1.5. Monoclonal antibodies

- Highly specific and identical antibodies made by identical B cell clones

Monoclonal antibody formation:

- A mouse is injected with relevant antigen, stimulating immune response
- Plasma cells specific to antigen are extracted from the spleen and fused with cancerous cells forming hybridoma cells
- Hybridoma cells that produce the required antibody are cloned

Diagnosis with monoclonal antibodies:

- Radioactive chemicals are attached to each antibody that binds to fibrin
- Radioactivity emitted by these antibodies can be detected by gamma rays camera, thus finding the position of a clot
- The same method can be used to locate cancer cells and identify the exact strain of a virus or bacterium during an infection

Treatment with monoclonal antibodies

- MABs from rats into humans trigger an immune response as they are non-self
- This is overcome by altering genes that code for polypeptide chains of antibodies into human sequences and the type/position of sugar groups into human antibodies
- Used in cancer therapy by marking cancerous cells for their destruction or binding to protein produced by T cells that reduces immune response
- Controls over/inappropriate production of B cells, preventing leukaemia and autoimmune diseases

1.6. Active and passive immunity

Immunity	Active	Passive
Antigen encountered	Yes	No
Immune response	Yes	No
Time before antibodies disappear in blood	1-2 weeks during immune response	Immediate
Production of memory cells	Yes	No
Protection	Permanent	Temporary
Origin	Antigen enters body so immune response occurs by infection (natural) or vaccines (artificial)	Gained through breast milk or placenta (natural) or through injected antitoxins (artificial)

- Natural active immunity: immune response after being infected by the pathogen
- Artificial active immunity: immune response after vaccination
- Natural passive immunity: mother's milk (colostrum), placenta
- Artificial passive immunity: injecting antibodies/antitoxins in the body

1.7. Vaccination

Vaccination:

- An antigenic material, which could be a live, dead or attenuated micro-organism, or perhaps a harmless form of a toxic (toxoid) or simply surface antigens
- This allows our immune system to produce the requisite B and T cells without actually suffering the disease, mimicking natural immunity (long-term)

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- Less effective vaccines do not mimic an infection as they are dead bacteria or viruses that do not replicate inside the body
- They need booster injections to stimulate secondary responses that give enhanced protection

Common barriers to vaccination:

- Poor response due to:
  - The defective immune system, so doesn't produce the necessary B and T cell clones
  - Suffer from malnutrition and don't have enough protein to make antibodies
- Antigenic variation:
  - Pathogens mutate rapidly and form different strains with different strains
  - Memory cells unable to recognize pathogens; hence vaccine is ineffective
  - Plasmodium is eukaryotic and has many more genes and antigens on its surface
- Antigenic concealment:
  - Pathogens escape from the immune system by living inside host cells or suppressing the immune system
  - Beyond the reach of antibodies
- Live virus and herd immunity:
  - People may also be injected with a live virus and could potentially pass it out in their faeces during the primary response, evolve by mutation in the environment, and infect others
  - This is why we vaccinate everyone at the same time, known as herd immunity
  - Herd immunity interrupts transmission in a population so that those who are susceptible never encounter the infectious agents concerned

Why can't we vaccinate against malaria?

- No effective vaccine
- Protoctists are eukaryotes, so many more genes and antigens than bacteria and viruses
- Display different antigens on its cell surface for different strains or different stages of its life cycle - shows antigenic variation
- Parasite changes antigens during infection
- Plasmodium hides in the liver, and RBC - antigenic concealment
- Needs more than one type of vaccine which targets common antigens present in all stages of the life cycle
- Needs more funding and research

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